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1. What are the different protocols you observe at the following layers of protocol stack?

After following the given instruction in the document i able to capture 16700 packet within 6 mins

i) Application layer:

In 16700 packets, found 497 (3.0%) of application protocol which include

- HTTP / JSON → contain 1 packet i.e less than 0.1%.
- DHCP → contains 19 packet i.e 0.1%.
- DNS → contains 2.8% i.e 462 packet.
- HTTP → contain 16 (0.1%) packet
- SSDP → contain 0.3% i.e 55 packets.
- Link-local multicast Name resolution → contain 2.0% i.e 8 packet in each IPv4 & IPv6.
- Multicast DNS → 9.5% i.e 430 packet in IPv6 & 2.5% i.e 420 packet in IPv4.
- Dynamic-host configuration protocol → contain 0.1% i.e 19 packet in IPv4 & 0.1% i.e 15 packets in IPv6.

ii) Transport layer:

- TCP → contain 38.6% i.e 6393 packets
- UDP → contain 26.2% i.e 4381 packets in IPv4 & 2.7% i.e 455 packet in IPv6

iii) Network layer:

- Internet group management protocol → contain 0.6% i.e 95 packet
- Internet control message protocol (ICMPv6) → contain 1.6% i.e 248 packet.

• IPv4 : contain 65.1% i.e 10865 packets.

• IPv6 : contain 4.2% i.e 703 packets

Q2 What is the total amount of data being received for the following two case?

1) NeverSSL.com → IP Add → [34.223.124.45]

Step 1 → filter IP address for specific website used IP.add
= 34.223.124.40.

Step: II → Statistics → conversation & then I found total
200 packet conversion b/w & total amount of data
received is 91 packet which is 32 KB

2) cornell.edu: IP Add → [128.253.173.244].

I found total 72 packet of 29 KB & total packet received
from cornell.edu is 22 KB of total packet 36.

Q3. Answer the following when you access the site
neverSSL.com →

3 → 1) What version of HTTP the server running?
→ only HTTP/1.1.

3 → 2) How many HTTP GET request do you observe? List
down the GET requests.

→ Total 6 HTTP GET request from neverSSL.com

* HTTP GET request are :-

i) 579 GET /online HTTP/1.1

ii) 580 GET /online HTTP/1.1

iii) 485 GET /favicon.ico HTTP/1.1

iv) 558 GET /online HTTP/1.1

v) 559 GET /online HTTP/1.1

vi) 495 GET /favicon.ico HTTP/1.1

∴ By using "http.request.method == "GET" and I.P. address 34.228.124.45

3 → 3. Identify TCP three way handshake used to establish the connection with ACK packets & write the sequence number of each of the three access.

sol Net request

Nb → 3295 → GET /online HTTP/1.1

Stream no → 101

Number	information	Sequence no
3194	SYN	0
3295	SYN, ACK	0
3296	ACK	1

Steps which i follow to answer:

S-1 → Get the detail of GET HTTP request
"http.request.method == "GET"

S-2 Select any request & check stream index from panel.

S-3 To get TCP handshake detail filter using this command →

"tcp.stream == 101"

S-4 ∴ Get the information from info column & non-column & panel for sequence no.